

ADD ARTICLE TITLE

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Abstract

ADD: research problem and its scale; objective; datasets and validation design; methods compared; principal quantitative results; main conclusion. Keep it factual, self-contained and within 250 words.

Keywords: ADD KEYWORD 1, ADD KEYWORD 2, ADD KEYWORD 3

1. Introduction

ADD: problem and motivation, including the scale of the data; why it matters in the application domain; gap in existing research; research questions (RQ1 to RQn); contributions; one sentence per section on the structure of the paper.

2. Related Work

ADD literature stream 1. ADD: directly relevant prior work and how it connects to this study.

ADD literature stream 2. ADD: directly relevant prior work and how it connects to this study.

ADD literature stream 3. ADD: directly relevant prior work; close with the gap this paper addresses.

3. Background

3.1. ADD Domain Background

ADD: requirements of the application domain that the evaluation must respect.

3.2. ADD Classical Methods Background

ADD: background on the established methods, with equations where the paper uses them, for example

$$\text{ADD FORMULA} \quad (1)$$

referenced as Eq. (1).

3.3. ADD Proposed Approach Background

ADD: formal problem statement, notation ($\mathcal{F}$, $\mathcal{D}_{\text{DEV}}$, $\mathcal{D}_{\text{OOT}}$, K) and the quantities on which methods are judged.

3.4. ADD Evaluation-Concept Background

ADD: background on reproducibility, drift or other evaluation concepts used later.

4. Methodology

4.1. Overall Research Pipeline

Figure 1 summarises the pipeline.

1. **ADD stage 1.** ADD description.
2. **ADD stage 2.** ADD description.
3. **ADD stage 3.** ADD description.

ADD: which stages use target information and which are target-free; what is frozen before held-out scoring; leakage controls.

4.2. Data Preprocessing

ADD: missing-value handling, scaling, categorical encoding, unseen-category handling, and the order of preprocessing and selection.

4.3. ADD Method Family 1

ADD: definition, implementation, criterion, hyperparameters and retained-feature rule for each method. A method inventory table:

4.4. ADD Method Family 2

ADD: model or prompt version, inputs, information contract, ranking or selection procedure, caching and reproducibility procedure.

*Corresponding author

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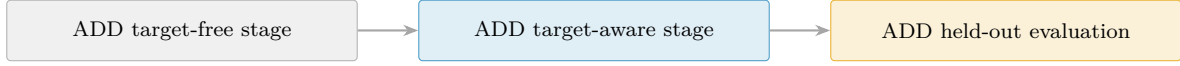


Figure 1: ADD PIPELINE FIGURE CAPTION. Grey boxes are target-free stages, blue boxes target-aware stages fitted on development data, orange boxes the locked held-out evaluation.

Table 1: ADD METHOD INVENTORY CAPTION.

Method	Family	Role	Labels
ADD METHOD	ADD FAMILY	ADD one-line description	yes/no
ADD METHOD	ADD FAMILY	ADD one-line description	yes/no

4.5. ADD Method Family 3

ADD: exact order of operations for each combined pipeline.

4.6. Development of the Protocol

ADD: protocol versions preceding the final evaluation and which changes moved the results; superseded values are labelled as such.

5. Experimental Setup

5.1. Datasets

ADD: for each dataset, source, observation unit, event definition, time coverage, sample sizes, event rates and feature universe.

5.2. Held-Out DEV/OOT Design

ADD: DEV and OOT construction, cutoff dates, fold construction, what is frozen before the held-out population is opened, and any change to the design made during the study.

5.3. Predictive Models

ADD: full configuration of each backbone, class weighting, solver, seed, iteration count and stopping rule.

5.4. Feature Budgets

ADD: how budgets were determined on development data only and when they were frozen.

5.5. Evaluation Metrics

ROC-AUC. ADD: definition and interpretation.

KS. ADD: definition, formula, direction of improvement.

Brier score. ADD: definition, formula, direction of improvement.

Capture@10 and Lift@10. ADD: definition, formula, direction of improvement.

PSI. ADD: formula, binning and reference construction, frozen-bin procedure.

Nogueira stability. ADD: formula, universe used in the denominator, aggregation procedure.

Jaccard similarity. ADD: formula and pairwise fold aggregation.

5.6. Statistical Inference

ADD: paired test, null hypothesis, confidence-interval procedure, significance level, number of prespecified comparisons and multiple-testing correction.

5.7. Leakage and Reproducibility Controls

ADD: leakage controls, seed policy, versioning of data, code and models, and any other safeguards.

6. Results

6.1. Primary OOT Comparison

ADD: the prespecified headline comparisons and their inference; separate observations from explanations.

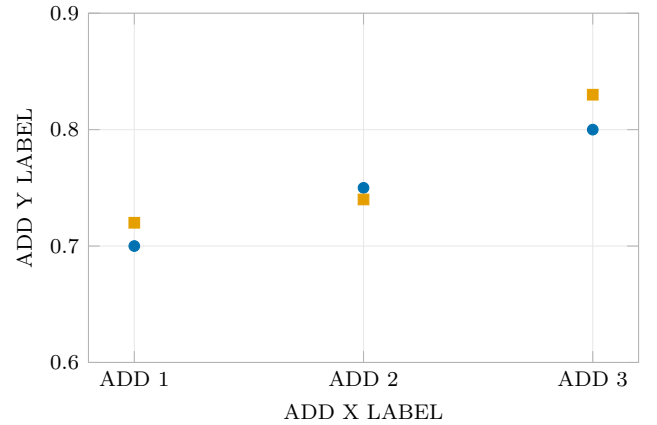


Figure 2: ADD FIGURE CAPTION.

6.2. Secondary Metrics

ADD: secondary metrics without duplicating the primary table.

6.3. ADD Further Result 1

ADD: numerical results and factual interpretation.

Table 2: ADD DATASET SUMMARY CAPTION.

	ADD DATASET 1	ADD DATASET 2	ADD DATASET 3
Observation unit	ADD	ADD	ADD
DEV window	ADD	ADD	ADD
DEV N (event rate)	ADD	ADD	ADD
OOT window	ADD	ADD	ADD
OOT N (event rate)	ADD	ADD	ADD
Feature universe	ADD	ADD	ADD

ADD table note if needed.

Table 3: ADD PRIMARY RESULTS CAPTION.

Case	ADD arm A (AUC)	ADD arm B (AUC)	Reference	Δ AUC [95% CI]	Holm p
ADD CASE	ADD	ADD	ADD	ADD	ADD

^a ADD table note if needed.

6.4. ADD Further Result 2

ADD: numerical results and factual interpretation.

Scalability and cost. ADD: runtime, calls and cost of the proposed stage relative to the baselines, labelled descriptive where measurement conditions differed.

7. Discussion

7.1. Main Findings

ADD: answers to RQ1 to RQn, in the order asked.

7.2. ADD Trade-off Subsection

ADD: supported trade-offs between the outcomes measured.

7.3. Cross-Dataset Behavior

ADD: which findings reproduce across datasets and models and which are dataset- or model-specific.

7.4. Limitations

ADD: limitations of data, models, methods, validation design, inference, reproducibility and generalisability.

8. Conclusion

ADD: the research question, the strongest supported findings, implications for practice and research, and next steps. Do not introduce new results.

CRedit authorship contribution statement

ADD AUTHOR 1: ADD roles. **ADD AUTHOR 2:** ADD roles.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Disclaimer

ADD: institutional disclaimer if an author is affiliated with a public body, otherwise delete this section.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

ADD: repository links or DOIs for datasets, code, cached artefacts and experiment outputs; state the exact reason for anything that cannot be shared.

- ADD DATASET: <https://ADD-URL>
- Code and experiment artefacts: <https://ADD-URL>

Declaration of generative AI and AI-assisted technologies in the writing process

ADD: tool and purpose, followed by the statement that the authors reviewed and edited the content and take full responsibility for it; delete if no such tool was used.

Appendix A. ADD APPENDIX A TITLE

ADD: one paragraph stating what is reproduced below and where the main text refers to it.

Appendix A.1. ADD listing title

ADD verbatim material (prompt template, configuration, feature list).

References

- [1] ADD: A. Author, Title, Journal Volume (Issue) (Year) pages. <https://doi.org/ADD>
- [2] ADD: B. Author, Title, in: Proceedings, Year, pp. 1–10.